

Lesson 2.7: Sports Science – Force, Motion & Energy

Newton's Laws, Kinetic Energy,
Inertia & More

Newton's First Law of Motion

- An object in motion stays in motion, and an object at rest stays at rest, unless acted on by a force.
- Also called the Law of Inertia.
- Example: A hockey puck gliding on ice keeps going until something stops it.

Newton's Second Law of Motion

- Force = Mass $\times$ Acceleration ($F = ma$).
- More mass = more force needed for same acceleration.
- Example: A bowling ball needs more force to accelerate than a baseball.

Newton's Third Law of Motion

- For every action, there is an equal and opposite reaction.
- Example: When you hit a ball with a bat, the bat also feels a force back from the ball.

Kinetic Energy

- Energy an object has because it's moving.
- Formula: $KE = \frac{1}{2} \times \text{mass} \times \text{velocity}^2$
- Example: A golf ball hit with a fast swing has high kinetic energy and goes farther.

Check Your Understanding!

Why do hockey players keep sliding after they stop pushing with their skates?

- A. Inertia
- B. They're on a slope
- C. Gravity pulls them forward
- D. They're still using their muscles

Check Your Understanding!

A baseball and a bowling ball are thrown with the same acceleration. Which required more force and why?

Check Your Understanding!

When a soccer player kicks a ball, what is the reaction force?

- A. The ball pushing back on the foot
- B. The player flying backwards
- C. The goal net shaking
- D. The wind resisting the ball

Check Your Understanding!

Two balls have the same mass, but we know that one has more kinetic energy. How do we know it? Use the formula to explain:

$$KE = \frac{1}{2} \times \text{mass} \times \text{velocity}^2$$